FastRobox 1.0 — Judges' Checklist

12 levels · 500 marks · one copy per team

Team ID 01		Team name	
Judge		Station	

How to use: the team calls you (at most four calls) and names the levels to judge — consecutive levels, starting with its next level not yet passed. Use the round sheet for those levels and do only the rows tagged with them, in order; one action may count for several levels. Tick a line (Att. 1 or Att. 2) only when you see it yourself. A level passes when all its lines are ticked in the same attempt; if a level fails, stop the call there. Once or twice per call, ask the team to repeat one "Video must show" item live (spot check); if it fails, that level's attempt fails. A time up to 0.5 s over a limit: measure once more, the better result counts. Two failed attempts close that level and all later levels. 2nd-attempt pass = 70% of the marks.

Calls used: ☐ 1 ☐ 2 ☐ 3 ☐ 4 Spot checks (level and item): _____

Round A — Levels 1–3 — about 10 min — Bring: Your phone with a 2.4 GHz hotspot; candle and lighter; fire extinguisher; stopwatch

Before you start: The Serial Monitor is open on the team's laptop at 115200 baud. Turn on your phone's hotspot (2.4 GHz; on iPhone turn on "Maximize Compatibility"). Keep it on until the round ends. Keep the fire extinguisher within reach.

Row	Do this	Lv	You must see	Att. 1	Att. 2
A1	Watch the light for 10 seconds while tapping the kit card 3 times, then hold the card on the reader for 5 seconds. Ask the team to type "health", then "tasks".	1	The light keeps a steady 2 Hz rhythm; exactly 3 "[CARD]" lines for the taps and 1 for the held card	☐	☐
		1	"health" gives every value listed for it in the Serial Commands table; "tasks" lists at least 3 tasks using both cores	☐	☐
A2	Ask the team to type "hang".	1	The board restarts by itself within 10 seconds, prints "[EVENT] BOOT reason=WATCHDOG" and starts with the door "LOCKED"	☐	☐
A3	Ask the team to type "reset". On the team's laptop, join "SmartClass-<TeamID>" with the password shown on the Serial Monitor, open 192.168.4.1 and type in your hotspot's name and password, a Room ID you choose and the team's device token. Then connect the laptop to your hotspot and open the device's IP address and the mDNS address.	2	Blue light; the access point asks for the password; the device joins your hotspot and prints its IP address and mDNS name	☐	☐
		2	Both addresses show the local status page with your Room ID and every field	☐	☐
A4	Tap the card. Ask the team to show the newest database records for 20 seconds, then to send an old message again (same "seq").	2	The local page updates within 2 seconds of the tap; records arrive every 5 ± 1 seconds with every field, and the tap is stored with its UID	☐	☐
		2	The resent message is rejected and the number of records does not change	☐	☐
A5	Hold the candle about 30 cm from the flame sensor for at least 3 seconds and, with the flame still there, tap the Teacher tag. Put the candle out; tap the student card, then the Teacher tag.	3	"FIRE": continuous buzzer, fast red flashing, door "UNLOCKED", stored in the database within 3 seconds	☐	☐
		3	Nothing silences the buzzer while the flame is there; after it has gone, only the Teacher tag does, and the door stays "UNLOCKED"	☐	☐
A6	Unplug the board for 40 seconds and plug it back in. Then ask the team to send Reset fire.	2	The platform shows offline within 30 seconds and online again afterwards, with a "BOOT" event; the device reconnects with no setup	☐	☐
		3	After the restart the door is still "UNLOCKED"; Reset fire makes the door "LOCKED" and the status Safe	☐	☐
A7	Hold the free metal end of the panic wire for 3 seconds.	3	"PANIC" (light, buzzer, local page), stored in the database within 3 seconds	☐	☐

After the round: The team may set the device up on its own network again.

Round B — Levels 4–8 — about 15 min — Bring: Your phone (camera); two browsers (laptop and phone); stopwatch

Before you start: Collect the judges' logins and the list of page addresses from the team. Check that the team has its three firmware files ready: a working one with a higher version number, a damaged copy, and a faulty one that cannot reach the server. Set-up by the team for Level 7: A and B are in the class held in this room; C is in another class; D is enrolled here but will not come; the card is registered to A; a Guardian is linked to B and D. Ask everyone not taking part in a row to stay out of the camera's view.

Row	Do this	Lv	You must see	Att. 1	Att. 2
B1	Log in as two roles you choose. As Teacher, keep the dashboard open, tap the card, and ask the team to type "health".	4	Each role sees only what it is allowed to see	☐	☐
		4	The tap shows within 5 seconds without reloading, and the device health panel matches the "health" output	☐	☐
B2	Touch the panic pad for 3 seconds and ask the team to show the Teacher's phone. Then, logged in as Student S2, open S1's attendance page address.	4	The Teacher's dashboard shows a red "PANIC" banner within 3 seconds, and the phone shows the message within 10 seconds	☐	☐
		4	"Not allowed" (401 or 403) and no data	☐	☐
B3	As Admin, set a Warning limit you choose between 25 °C and 35 °C.	5	The local status page shows the new limit within 10 seconds	☐	☐
B4	As Admin, upload the team's working firmware file with the higher version number, and wait.	5	Within 3 minutes the device restarts with the new version on the Serial Monitor, the local page and the dashboard health panel; the new limit and the Wi-Fi details are kept	☐	☐
B5	Open the dashboard in two browsers as two different roles. As Admin, add a Danger TEST notice; then remove it.	6	Both dashboards show the weather, the 24-hour forecast and the red banner within 10 seconds; within 10 seconds the device glows purple, beeps three times and its local page shows "WEATHER ALERT" with the text	☐	☐
		6	Removing the notice clears the banner and the device alert	☐	☐
B6	Open today's daily report. Then, as Teacher, ask the assistant one question you choose about today's stored data.	6	The report shows today's minimum, maximum and average and the number of alarms of each type, including the fire and panic from Round A	☐	☐
		6	The assistant's answer matches the report	☐	☐
B7	As Admin, set the start of the class in this room to now. Tap the card with nobody facing the camera; then A taps it while facing the camera.	7	Denied with "NO FACE" within 4 seconds, decided by the device itself ("NO_FACE" on the Serial Monitor)	☐	☐
		7	A gets in within 5 seconds and is marked Present; the log of received images shows an image smaller than the full camera frame, of the size printed on the Serial Monitor	☐	☐
B8	As Admin, set the class start to 11 minutes ago and register the card to B; B checks in.	7	B is marked Late and the Guardian gets the Late notice within 60 seconds	☐	☐
B9	Register the card to A again. B taps it while facing the camera. Then take a photo of A with your phone and show it to the camera on your phone screen while tapping the card, for 10 seconds.	8	Denied with "FACE MISMATCH"; the Teacher sees an alert with a snapshot within 10 seconds	☐	☐
		8	The phone photo never unlocks the door	☐	☐

After the round: Ask the team to set the Warning limit back to 32 °C.

Round C — Levels 9–12 — about 12 min — Bring: Two chairs; masking tape; your phone; stopwatch; paper for the trial orders

Before you start: Place two chairs in seat zones 1 and 2 of the camera's "ROOM" view. Check that the 2 × 2 floor grid (about 1 m per zone, numbered 1–4) marked before the clock started is still in place. The device is "LIVE" on the team's own hotspot. Ask everyone not taking part in a row to stay out of the camera's view and outside the grid.

Row	Do this	Lv	You must see	Att. 1	Att. 2
C1	As Admin, set the capacity to 1 and select "ROOM" (allow 1 minute to re-aim). Wait up to 6 seconds after each of these: view empty; one person sits in seat zone 2; a second person stands in view outside the seat zones.	9	The count and seat map are right each time: 0 / none; 1 / zone 2; 2 / zone 2 only	☐	☐
		9	"OVER CAPACITY" appears on the dashboard and the local page only while 2 people are in view	☐	☐
C2	Open the live view as Teacher and count its updates for 10 seconds while walking in front of the camera.	9	At least 20 updates in 10 seconds, with a box following you	☐	☐
C3	Write down a random order of 4 trials and run them with a team member in seat 1 or 2: two hands raised for 3 seconds, and two other movements (touch head; stretch arms sideways).	10	At least 3 of the 4 trials are correct	☐	☐
		10	Each detected raise appears within 5 seconds, on the correct seat, once, and the per-seat count goes up by one	☐	☐
C4	Cover the lens with masking tape. Everyone stands outside the floor grid, at least 2 m away, and keeps still for 10 seconds.	9	"COVERED" within 6 seconds, with a Teacher alert, a snapshot and the cyan light	☐	☐
		11	"EMPTY" within 5 seconds, staying "EMPTY" for the whole 10 seconds	☐	☐
C5	Write down a random order of the 4 grid zones. A judge walks into the grid and stands in each zone in that order for 10 seconds (the zone shown at the end of the 10 seconds counts), then walks out to at least 3 m away.	11	"OCCUPIED" within 5 seconds of entering; at least 3 of the 4 zones are correct, and each time the dashboard map highlights the zone shown on the local page within 3 seconds	☐	☐
		11	"EMPTY" within 5 seconds of leaving	☐	☐
C6	Remove the tape from the lens.	9	The camera is back to "OK" within 6 seconds	☐	☐
C7	As Admin, set the start of the class in this room to now, with A in it and the card registered to A. Ask the team to switch its hotspot off for 90 seconds. After 40 seconds, join "SmartClass-<TeamID>" on your phone and open 192.168.4.1; ask the team to show a BLE scanner on its own phone. A taps the card; then tap the Teacher tag.	12	"OFFLINE" on the Serial Monitor and the local page, with the waiting-records count rising; "SC-<TeamID>-OFFLINE" in the BLE scanner	☐	☐
		12	A is let in (solid green); the Teacher tag does not unlock the door	☐	☐
C8	At 90 seconds, ask the team to switch the hotspot on. Wait 60 seconds and ask for the outage records.	12	"SYNCING", then "LIVE"; every outage record arrives once with its original time; A is Present at the tap time, flagged face-not-verified	☐	☐

Evidence Check (after judging closes, for every level passed live)

For each level, also check: video at least 720p, team ID and level stated, at most 10 minutes; write-up at most two pages with the five headings and screenshots of every surface the level uses; Git tag level-NN with commit hash; uploaded within 30 minutes. Any missing item means the level scores zero.

Level 1: Dual-Core Tasks, Health and Watchdog			
L1	Video	The DHT11 data wire and the RC522 SDA wire unplugged for 15 seconds: "SENSOR ERROR" for the RFID reader within 3 seconds and for the DHT11 within 12 seconds, with magenta light and no freeze; both clear after reconnecting	☐
	Write-up	A "[STATUS]" line every second in the fixed format	☐
	Write-up	Read periods: DHT11 5 s, flame 100 ms, RFID 200 ms	☐
	Write-up	Flame at start-up gives door "UNLOCKED"	☐
Level 2: Wi-Fi Setup and Secure Telemetry			
L2	Video	The hotspot switched off for 20 seconds: the device reconnects within 30 seconds without restarting (uptime keeps counting)	☐
	Video	The BOOT button held for 5 seconds: the access point returns; set up with a wrong hotspot password, the device is back in setup mode within 15 seconds and the page shows an error	☐
	Video	The backend stopped for 20 seconds: records resume within 30 seconds without resetting the device	☐
	Write-up	HTTPS or MQTT over TLS with the server's certificate checked	☐
	Write-up	No password, key or token in the repository; the README explains how to build and run the system	☐
Level 3: Heat, Fire, Panic and Fail-Safe Door			
L3	Video	Warm air from a hair dryer on the DHT11 until Danger, then cooling: yellow, then red with a beep every 2 seconds; heat index and trend shown; no "FIRE" and no change to the door; exactly one Danger → Warning and one Warning → Safe change, with Safe only at 30 °C or below	☐
	Video	A candle flicked past the flame sensor for about 1 second, and a touch on the panic pad shorter than 1 second: nothing happens	☐
	Video	Gentle warming: the trend shows Rising with no change to the light, the buzzer or the door	☐
	Video	The DHT11 unplugged during Warning: "quality" shows "bad" and the status does not fall to Safe	☐
	Write-up	The touch-pad baseline is calibrated at start-up	☐
Level 4: Secure Platform, Roles and Alerts			
L4	Video	All four roles logged in, each seeing only what it is allowed to see	☐
	Video	The dashboard switched to Bangla: all menus and labels in Bangla on two pages, then in English again	☐
	Video	5 wrong passwords in a row: the correct password is refused for 30 seconds, then accepted; the login history lists the attempts with their times	☐
	Write-up	Passwords stored hashed (bcrypt or argon2)	☐
	Write-up	A Teacher-only page opened by a Student, and another child's page opened by the Guardian, are refused with no data	☐
Level 5: Remote Settings, OTA and Rollback			
L5	Video	The DHT11 warmed past the new limit: Warning (yellow) appears after 3 seconds at or above it	☐
	Video	A Warning limit of 45 °C refused with a clear message	☐
	Video	The board switched off and on: the changed limit is still shown and used	☐
	Video	The damaged firmware file rejected with the device keeping its current version, and the faulty firmware installing, then returning by itself to the previous version with an "OTA_ROLLBACK" event	☐
	Write-up	A Teacher cannot open the settings page; the change log shows who changed what and when	☐
Level 6: Weather, History and AI Assistant			
L6	Video	The weather service blocked: the board shows the last data with its age and "Data stale"	☐
	Video	The chart with the heat peak, the event history filtered to "FIRE" and "PANIC" with their times, and its CSV export matching the page	☐
	Video	An assistant question about a date with no data, answered with "no data" and nothing invented	☐
	Write-up	Students and Guardians cannot open the history, report or assistant pages	☐
	Write-up	The weather service key never reaches the browser	☐
Level 7: Card and Face Check-In			
L7	Video	A second check-in by A: the door opens but no record is added	☐
	Video	The card registered to C (another class): "NOT IN THIS CLASS"; the card unregistered: "UNKNOWN CARD"	☐

	Video	The class start set to 31 minutes ago: D is marked Absent and the Guardian is notified within 60 seconds; A stays Present, B stays Late, and the Guardian sees only B and D	☐
	Write-up	Consent notice, at least 3 face images per student, embeddings stored and photos deleted	☐
	Write-up	A tap with no class running is denied with "NO CLASS NOW"	☐
Level 8: Spoofing, Lockout and Data Deletion			
L8	Video	B tapping A's card until the third mismatch: "CARD BLOCKED", even for A; after the Admin unblocks it, A gets in	☐
	Video	Two Teacher windows pressing Ack on one alert at the same moment: exactly one acknowledgment, naming who and when	☐
	Video	Student C deleted by the Admin: no face data, card link or snapshot of C remains; the audit log shows who deleted C and when	☐
	Write-up	A Student cannot open a snapshot's address	☐
	Write-up	The card stays blocked for 5 minutes unless the Admin unblocks it	☐
Level 9: Occupancy, Live View, Night Watch and Tamper			
L9	Video	Night mode, with a person in front of the camera for 5 seconds: an "INTRUDER" alert with a snapshot reaches the Admin within 10 seconds, and the local page shows "INTRUDER"	☐
	Video	The camera turned about 45° away: "MOVED" within 6 seconds, and "OK" within 6 seconds after turning it back	☐
	Video	A card tapped while the camera is covered at "DOOR": denied with "CAMERA FAULT"; a card tapped at "ROOM": denied with "CAMERA NOT AT DOOR"	☐
	Write-up	Counting and live-view frames are not stored (only the "INTRUDER" snapshot is kept); Students cannot open the live view	☐
	Write-up	People walking through the view cause no camera fault; the camera-health scores are shown live	☐
Level 10: Own-Trained Hand-Raise Detector			
L10	Video	The Model page: at least 100 images per class, at least 20% held out, held-out accuracy, confusion matrix and time per prediction	☐
	Write-up	The training code is in the repository; no ready-made hand-raise model is used	☐
Level 11: Wi-Fi CSI Presence and Localization			
L11	Video	The "csi" command showing at least 20 CSI packets per second, and the Model page with samples per zone and empty, held-out accuracy and confusion matrix	☐
	Write-up	The camera is not used for any decision; at least 20% of the CSI samples are held out	☐
Level 12: Offline, Bluetooth and Power-Cut Recovery			
L12	Video	The board switched off for 10 seconds during an outage: it keeps working offline, the waiting-records count is kept, and the records from before the cut are later sent once with their original times	☐
	Video	The device put in a non-default state (for example the fire state latched) and the power cut twice in a row: after each restart the light, local page, door and dashboard agree within 20 seconds	☐
	Write-up	At least 30 minutes of telemetry can be buffered; a half-written record is discarded	☐